Butterfly Effects of SGD Noise:
Error Amplification in Behavior Cloning and Autoregression

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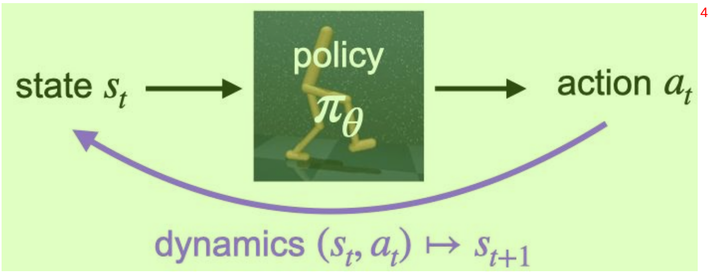
Behavior cloning & feedback loops [3]



[4]

BC: fit θ to a collection of demonstrations from expert policy $\pi^\star$, minimizing 1-step loss [5]

(e.g.) $\ell_{\mathrm{BC}}(\theta) = \mathbb{E}_{\text{expert's } (s_t, a_t)} \left[\|\pi_\theta(s_t) - \pi^\star(s_t)\|^2 \right]$ [6]

Usual view: BC reduces offline imitation learning to supervised learning [7]

$\ell_{\mathrm{BC}}(\theta) \le \varepsilon$ [8] → $J_H(\pi_\theta) \ge J_H(\pi^\star) - C \cdot \varepsilon$ [10]

train on 1-step loss [9] → evaluate multi-step rollout rewards [11]

Q1: *When BC works, how big is C in practice?* [12]

Q2: *When BC fails, must we improve the data? Or can we improve upon standard training?*

Special case: pretraining vs. generation in LLMs [13]

- s_t : sequence of t tokens
- a_t : append a single token
- π_θ : autoregressive language model [14]

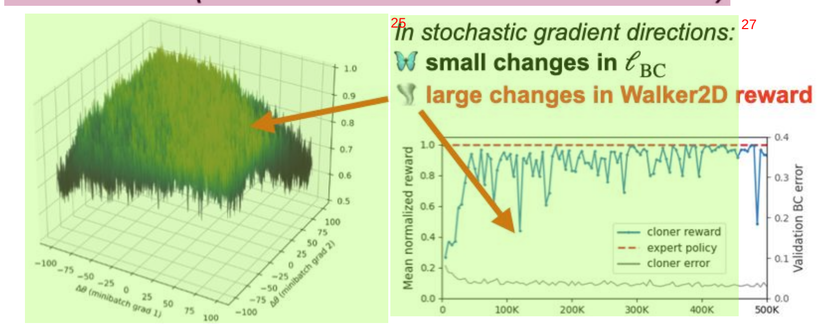
Gradient Variance Amplification [15]

A1: *Long-horizon rollouts can be extremely sensitive to small SGD fluctuations.* [16]

GVA: an empirical pathology in BC with NNs [17]

- [18] *small changes in θ due to minibatch noise* [19]
- [20] *...causing small changes in 1-step $\ell_{\mathrm{BC}}(\theta)$...* [21]
- [22] *...and **chaotic** long-rollout rewards $J_H(\pi_\theta)$* [23]

Evidence: *(across locomotion tasks & archs)* [24]



[25]

fractal reward landscape [26] | oscillations during training [28]

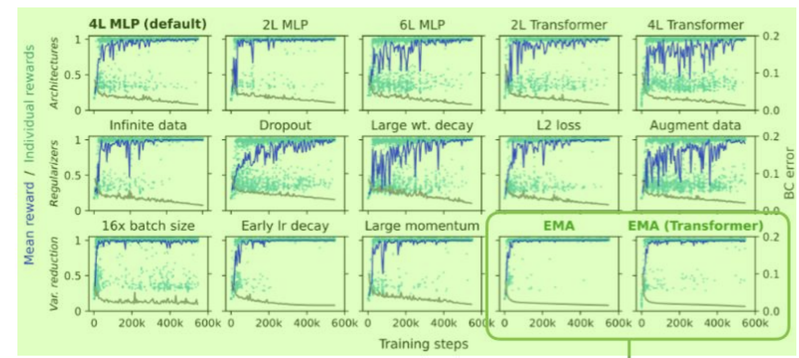
(in the paper: accompanying quantitative studies) [29]

Theory: vignettes in toy models [30]

- marginally stable linear dynamics, linear $\pi, \pi^\star$ [31]
- "CliffLQR": halfspace reward → GVA
- SDE limit: EMA & LR schedules both work

Algorithmic mitigations for GVA [32]

A2: *Variance reduction techniques mitigate the oscillations; EMA is highly effective.* [33]



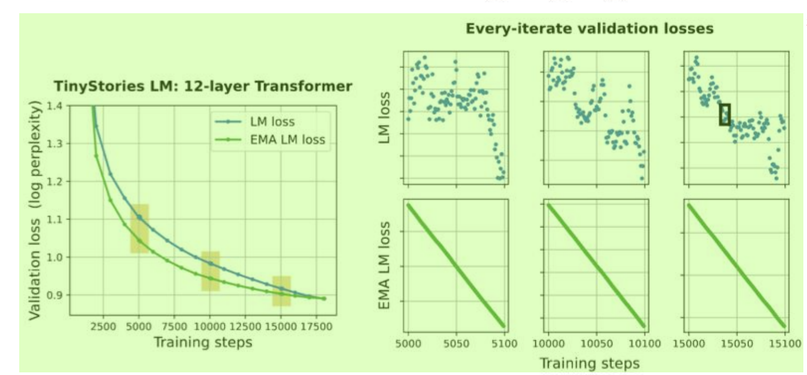
[34]

Exponential Moving Average [35]

$\theta_{t+1} \leftarrow \theta_t - \eta_t \widetilde{\nabla}_\theta \ell_{\mathrm{BC}}(\theta)$ (or Adam, etc.) [36]

$\bar{\theta}_{t+1} \leftarrow (1-\beta_t)\bar{\theta}_t + \beta_t \cdot \theta_t$, use $\bar{\theta}_t$ for inference

Bonus: GVA & EMA in language generation [37]



[38]

EMA improves & stabilizes LM iterates before learning rate cooldown [39]